

PYQ

Winter 2018

Q.01

DBATU
M-3



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Q.01.(a)

Show that, $\int_0^{\infty} \frac{\sin at}{t} dt = \frac{\pi}{2}$

04 Marks

Solution :

→ To show that : $\int_0^{\infty} \frac{\sin at}{t} dt = \frac{\pi}{2}$

→ Let, $I = \int_0^{\infty} \frac{\sin at}{t} dt$

Step(01) – To set given integration as a Definition of L.T.

$$\therefore I = \int_0^{\infty} 1 \times \frac{\sin at}{t} dt$$

• **Use** : $e^{-0 \cdot t} = e^0 = 1 \rightarrow 1 = e^{-0 \cdot t}$

$$\therefore I = \int_0^{\infty} e^{-0 \cdot t} \times \frac{\sin at}{t} dt \quad \dots (1)$$

Step(02) – Compare equation (1) with Definition of L.T.

→ We know, Defⁿ of L.T. is $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt \quad \dots (2)$

→ Now, we compare equation (1) and (2), we get :

$$\left\{ \begin{array}{ll} (1) \ I = L[f(t)] & \dots (M) \\ (2) \ s = 0 & \dots (N) \\ (3) \ f(t) = \frac{\sin at}{t} & \dots (O) \end{array} \right.$$

Step(03) – We have to find value of Given Integration (i.e. Value of I).

$$\therefore I = L[f(t)] \quad \dots [\because \text{from (M)}]$$

→ But, $f(t) = \frac{\sin at}{t}$ [$\because$ from (O)]

$$\therefore I = L \left[\frac{\sin at}{t} \right] \text{(P)}$$

Step (04) – Now, to find $L \left[\frac{\sin at}{t} \right]$.

* First Location :

$$\therefore L(\sin at) = \frac{a}{s^2 + a^2}$$

* Second Location :

→ By theorem - Division by t

$$\begin{aligned} \therefore L \left(\frac{\sin at}{t} \right) &= \int_s^\infty \frac{a}{s^2 + a^2} ds \\ &= a \times \int_s^\infty \frac{1}{s^2 + a^2} ds \end{aligned}$$

• Use : $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right)$

Here, $x = s$ & $a = a$

$$= a \times \left[\frac{1}{a} \tan^{-1} \left(\frac{s}{a} \right) \right]_s^\infty$$

$$= a \times \left\{ \left[\frac{1}{a} \tan^{-1} \left(\frac{\infty}{a} \right) \right] - \left[\frac{1}{a} \tan^{-1} \left(\frac{s}{a} \right) \right] \right\}$$

• Use : $\left(\frac{\infty}{a} \right) = \infty$

$$= a \times \left\{ \frac{1}{a} \tan^{-1}(\infty) - \frac{1}{a} \tan^{-1} \left(\frac{s}{a} \right) \right\}$$

$$= \cancel{a} \times \frac{1}{\cancel{a}} \left\{ \tan^{-1}(\infty) - \tan^{-1}\left(\frac{s}{a}\right) \right\}$$

• Use : $\tan^{-1}(\infty) = \frac{\pi}{2}$

$$= \frac{\pi}{2} - \tan^{-1}\left(\frac{s}{a}\right)$$

• Use : $\tan^{-1}(\theta) + \cot^{-1}(\theta) = \frac{\pi}{2} \rightarrow \cot^{-1}(\theta) = \frac{\pi}{2} - \tan^{-1}(\theta)$

$\rightarrow$ Here, $\theta = \left(\frac{s}{a}\right)$

$$\therefore L\left(\frac{\sin at}{t}\right) = \cot^{-1}\left(\frac{s}{a}\right)$$

$\rightarrow$ Put, this value in equation (P)

$$\therefore I = \cot^{-1}\left(\frac{s}{a}\right)$$

Step (05) – Put value of S, from equation (N).

$$\therefore I = \cot^{-1}\left(\frac{0}{a}\right)$$

$$\therefore I = \cot^{-1}(0)$$

• Use : $\cot^{-1}(0) = 90^0 = \frac{\pi}{2}$

$$\therefore I = \frac{\pi}{2} \dots (\text{Ans})$$

Q.01.(b)**Find the Laplace transform of** $\int_0^t \frac{e^{-3u} \sin 2u}{u} du$ **04 Marks****Solution :**→ To find the Laplace transform of $\int_0^t \frac{e^{-3u} \sin 2u}{u} du$ → i.e. to find $L \left[\int_0^t \frac{e^{-3u} \sin 2u}{u} du \right]$ → Here, we replace u by t , for convenient of solution.

→ So, to find $L \left[\int_0^t \frac{e^{-3t} \sin 2t}{t} dt \right]$

(Since, we know, property of definite integration, $\int_a^b f(x) dx = \int_a^b f(t) dt$)

*** First Location :**

$$\therefore L(\sin 2t) = \frac{2}{s^2 + 2^2} \quad \dots \left[\because L(\sin at) = \frac{a}{s^2 + a^2} \right]$$

$$\therefore L(\sin 2t) = \frac{2}{s^2 + 4}$$

*** Second Location :**→ By theorem - Division by t

$$\therefore L\left(\frac{\sin 2t}{t}\right) = \int_s^\infty \frac{2}{s^2 + 4} ds$$

$$= 2 \times \int_s^\infty \frac{1}{s^2 + 2^2} ds$$

• **Use :** $\int \frac{1}{x^2 + a^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$

Here, $x = s$ & $a = 2$

$$= 2 \times \left[\frac{1}{2} \tan^{-1} \left(\frac{s}{2} \right) \right]_s^{\infty}$$

$$= 2 \times \left\{ \left[\frac{1}{2} \tan^{-1} \left(\frac{\infty}{2} \right) \right] - \left[\frac{1}{2} \tan^{-1} \left(\frac{s}{2} \right) \right] \right\}$$

• Use : $\left(\frac{\infty}{2} \right) = \infty$

$$= 2 \times \left\{ \frac{1}{2} \tan^{-1}(\infty) - \frac{1}{2} \tan^{-1} \left(\frac{s}{2} \right) \right\}$$

$$= \cancel{2} \times \frac{1}{\cancel{2}} \left\{ \tan^{-1}(\infty) - \tan^{-1} \left(\frac{s}{2} \right) \right\}$$

• Use : $\tan^{-1}(\infty) = \frac{\pi}{2}$

$$= \frac{\pi}{2} - \tan^{-1} \left(\frac{s}{2} \right)$$

• Use : $\tan^{-1}(\theta) + \cot^{-1}(\theta) = \frac{\pi}{2} \rightarrow \cot^{-1}(\theta) = \frac{\pi}{2} - \tan^{-1}(\theta)$

$\rightarrow$ Here, $\theta = \left(\frac{s}{2} \right)$

$$\therefore \boxed{L \left(\frac{\sin 2t}{t} \right) = \cot^{-1} \left(\frac{s}{2} \right)}$$

* Third Location :

$\rightarrow$ By theorem - First Shifting

$$\therefore L \left(e^{-3t} \times \frac{\sin 2t}{t} \right) = \left[\cot^{-1} \left(\frac{s}{2} \right) \right]_{s \rightarrow s+3}$$

$$\therefore \boxed{L \left(\frac{e^{-3t} \cdot \sin 2t}{t} \right) = \cot^{-1} \left(\frac{s+3}{2} \right)}$$

* Fourth Location :

→ By theorem - L.T. of Integration

$$\therefore \mathcal{L} \left(\int_0^t \frac{e^{-3t} \cdot \sin 2t}{t} dt \right) = \frac{1}{s} \times \cot^{-1} \left(\frac{s+3}{2} \right)$$

→ But, in starting of solution,

we assume, variable u as t for convenient of solution.

→ So, here, in above equation, we replace variable t by u .

$$\therefore \mathcal{L} \left(\int_0^t \frac{e^{-3u} \cdot \sin 2u}{u} du \right) = \frac{1}{s} \times \cot^{-1} \left(\frac{s+3}{2} \right) \dots (\text{Ans})$$

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Q.01.(c)

Find the Laplace transform of the function

$$f(t) = \begin{cases} 2, & 0 < t < \pi \\ 0, & \pi < t < 2\pi \\ \sin t, & t > 2\pi \end{cases}$$

04 Marks

Solution : → Given that :

$$[1] \text{ Function : } f(t) = \begin{cases} 2, & 0 < t < \pi \rightarrow 0 \text{ to } \pi \\ 0, & \pi < t < 2\pi \rightarrow \pi \text{ to } 2\pi \\ \sin t, & t > 2\pi \rightarrow 2\pi \text{ to } \infty \end{cases}$$

[2] To find L.T. of the given function.

→ Here, we use Defⁿ of L.T.

$$\rightarrow \text{Def}^n \text{ of L.T. is } L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

→ This limit of integration (0 to ∞), we split as (0 to π) + (π to 2π) + (2π to ∞).

$$\therefore L[f(t)] = \int_0^{\pi} e^{-st} \cdot f(t) dt + \int_{\pi}^{2\pi} e^{-st} \cdot f(t) dt + \int_{2\pi}^{\infty} e^{-st} \cdot f(t) dt$$

$$= \int_0^{\pi} e^{-st} \cdot 2 dt + \int_{\pi}^{2\pi} e^{-st} \cdot 0 dt + \int_{2\pi}^{\infty} e^{-st} \cdot \sin t dt$$

$$= 2 \times \int_0^{\pi} e^{-st} dt + 0 + \int_{2\pi}^{\infty} e^{-st} \sin t dt$$

$$\therefore L[f(t)] = 2 \times A + B \dots (P)$$

→ Now, to find A & B separately.

$$[1] A = \int_0^{\pi} e^{-st} dt$$

• Use : $\int e^{px} dx = \frac{e^{px}}{p}$

→ Here, $p = -s$

$$= \left[\frac{e^{-st}}{-s} \right]_0^{\pi}$$

$$= \left[\frac{e^{-s\pi}}{-s} \right] - \left[\frac{e^0}{-s} \right]$$

• Use : $e^0 = 1$

$$\therefore A = -\frac{e^{-\pi s}}{s} + \frac{1}{s}$$

[2] $B = \int_{2\pi}^{\infty} e^{-st} \sin t \, dt$

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• Use : $\int e^{ax} \sin(bx) dx = \frac{e^{ax}}{a^2 + b^2} [a \sin(bx) - b \cos(bx)]$

→ Here, $a = -s$, $b = 1$, $x = t$

$$= \left\{ \frac{e^{-st}}{(-s)^2 + 1^2} [-s \cdot \sin(1 \cdot t) - 1 \cdot \cos(1 \cdot t)] \right\}_{2\pi}^{\infty}$$

$$= \left\{ \frac{e^{-st}}{s^2 + 1} [-s \sin(t) - \cos(t)] \right\}_{2\pi}^{\infty}$$

$$= \left\{ \frac{e^{-s \times \infty}}{s^2 + 1} [-s \sin(\infty) - \cos(\infty)] \right\} - \left\{ \frac{e^{-s \times 2\pi}}{s^2 + 1} [-s \sin(2\pi) - \cos(2\pi)] \right\}$$

• Use : (1) $e^{-s \times \infty} = e^{-\infty} = 0$

(2) $\sin(2\pi) = 0$

(3) $\cos(2\pi) = 1$

$$= \{0\} - \left\{ \frac{e^{-2\pi s}}{s^2 + 1} [0 - 1] \right\}$$

$$\therefore B = \frac{e^{-2\pi s}}{s^2 + 1}$$

→ Put, A and B in equation (P).

$$\therefore L[f(t)] = 2 \times \left[-\frac{e^{-\pi s}}{s} + \frac{1}{s} \right] + \frac{e^{-2\pi s}}{s^2 + 1}$$

$$\therefore L[f(t)] = \frac{-2e^{-\pi s}}{s} + \frac{2}{s} + \frac{e^{-2\pi s}}{s^2 + 1} \dots (\text{Ans})$$

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